

Ethan R. Partida

CONTACT

INFORMATION

School of Mathematics
Brown University
Providence, RI 02903

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INTERESTS

Algebraic Combinatorics, Algebraic Geometry, Tropical Geometry, Combinatorial Optimization

EDUCATION

Brown University, Providence, Rhode Island USA
Ph.D. student in Mathematics, Expected Spring 2027
Advisors: Melody Chan and Caroline Klivans

University of Minnesota, Minneapolis, Minnesota USA
Bachelor's degree in Mathematics, May 2022
Bachelor's degree in Computer Science, May 2022

PAPERS

The Chow Polynomial of the Noncrossing Partition Lattice (with J. Chun, N. Hayashida, and Z. Wang) [Submitted](#)

Hodge theory for combinatorial projective bundles (with M. Larson) [Submitted](#)

Algebraic Cobordism rings of Wonderful Varieties and Matroids (with R. Gandhi) [Submitted](#)

Graded Ehrhart Theory of Unimodular Zonotopes (with C. Crowley) [Submitted](#)

Line Shellings of Geometric Lattices (with S. Backman, G. Dorpalen-Barry, A. Nathanson, and N. Prime) [Preprint \(Revisions in progress, Fall 2026\)](#)

Shifted and threshold matroids. [Discrete Applied Mathematics, Volume 378, 2026](#)

On the topology of conormal complexes and posets of matroids (with A. Nathanson). [SIAM Journal on Discrete Mathematics, Volume 40, Issue 1, 2026](#)

Playing Cribbage with Reinforcement Learning and Minimax. [University of Minnesota Conservancy](#)

Computing Rational Powers of Monomial Ideals. (with P. Dongre, B. Drabkin, J. Lim, E. Roy, D. Ruff, A. Seceleanu, and T. Tang) [Journal of Symbolic Computation, Volume 116, 2023](#)

AWARDS

2025-2027 Simons Dissertation Fellowship

2025,2026 ×2 GAANN Graduate Fellowship Recipient

2024 NSF Graduate Research Fellowship Honorable Mention

2021 MAA Outstanding Student Paper Presentation

2021 JMM Honorable Mention Poster

2020 Ronald E. McNair Scholar

2020 Prof. Hans H. Dalaker Mathematics Undergraduate Scholarship

2020 William H. Burgum College of Science and Engineering Scholarship

2020 Texas State Rapid Fire Research Winner
 2019-2020 $\times 2$ National Hispanic Scholarship Fund Recipient
 2019 MN Space Grant Recipient
 2019 3M Diversity Scholarship

ADVISING AND MENTORING

Brown University, Summer UTRA Advisor of Thea Miller-Priddy, Haoyang Xu, Patrick Whiteford, and Zakhar Pysarevskyi (with Melody Chan), Summer 2026
 Brown University, Undergraduate Senior Thesis Advisor of Natsuka Hayashida (with Bena Tshishiku), Fall 2025-Spring 2026
 Brown University, Summer UTRA Advisor of Joseph Chun, Natsuka Hayashida, and Zhixing Wang (with Melody Chan), Summer 2025
 Brown University, Undergraduate Senior Thesis Advisor of Ameer Muhialdeen (with Melody Chan), Fall 2024-Spring 2025
 $\times 6$ Brown University, DRP Mentor, Spring 2023-Current
 University of Minnesota, [Mathematics Project at Minnesota](#), Peer Mentor, Fall 2021

PROFESSIONAL ACTIVITIES

Referee for the journal [Algebraic Combinatorics](#), 2026
 AMS Eastern Sectional Meeting, Organizer of the [Special Session on Group Actions and Combinatorics in Algebraic Geometry and Commutative Algebra](#), Spring 2026
 Brown DRP, [DRP Organizer](#), Fall 2024-Current
 Brown Horizons, [Reading Group Organizer](#), Spring 2023
 OURFA²M², [Experiences Panel Panelist](#), Fall 2022
 University of Minnesota, [Undergraduate Mathematics Research Seminar](#), Organizer, Fall 2021
 University of Minnesota, [SMART Learning Commons](#), Peer Tutor, Summer 2020-2022
 University of Minnesota, [North Star STEM Alliance](#), Member, Fall 2019-2022

SELECTED PRESENTATIONS

2026 University of Washington Combinatorics and Geometry Seminar
 2026 UC Berkeley Combinatorics Seminar
 2026 20 Minute Talk, Special Session on Algebraic and Geometric Aspects of Matroid Theory, AMS Fall Central Sectional Meeting
 2026 Dartmouth Combinatorics Seminar
 2026 University of Toronto Combinatorics Seminar
 2026 North Carolina State University Algebra and Combinatorics Seminar
 2026 Lightning Talk, Algebra and Geometry of Matroids Workshop, Casa Matemática Oaxaca
 2026 MIT/Harvard Richard P. Stanley Seminar in Combinatorics

2026 Poster Session, Formal Power Series and Algebraic Combinatorics (FPSAC)
 2026 University of Oregon Combinatorics Seminar
 2026 Graduate Online Combinatorics Colloquium
 2026 UCSD Combinatorics Seminar
 2026 University of Minnesota Combinatorics Seminar
 2026 20 Minute Talk, Commutative Algebra Meets Algebraic Combinatorics (CAAC)
 2024 20 Minute Talk, VIII Encuentro Colombiano De Combinatoria
 2024 Brown University Combinatorics Seminar
 2024 20 Minute Talk, Santander Workshop on Geometric and Algebraic Combinatorics
 2022 PME Poster Session, Joint Mathematics Meetings
 2021 Lightning Talk, Texas State Graduate Expo
 2021 Student Talk, Young Mathematicians Conference,
 2021 Student Paper Session, MAA Mathfest
 2021 Student Presentation, Brown University's SUMS
 2021 MAA Poster Session, Joint Mathematics Meetings

TEACHING

Brown University Teaching Assistant
 MATH 90 Calculus I, Fall 2023
 MATH 90 Calculus I, Spring 2024